Experiments :- Study of LAN Topology Network device and Address configuration

Objective :-

1) To study the LAN topology used in college Campus.

2) To identify networking device & understand their function

3) To analyze wired & wireless communication method used in the network.

4) To measure speed of Network for wired & wireless connections.

5) To obtained MAC, IP Address & subnet mask.

→ Problem statement - 1

- Draw & identify LAN topology of college campus & clearly label all components such as router & Switch & etc.

Solve :- The college campus network observed this study Consists of

* Layer - 3 Core switches in the server room
* Managed by PoE b Switches
* Wireless Access Point (APs)
* Ethernet port in classroom & Laboratories

→ **Network topology**

Hybrid Hierarchical topology which combines :

1) Tree topology
2) Extended star topology
3) Ring topology (server room)

→ **The network hierarchical layer**

① Core layer
② Distribution layer
③ Access layer

① **Core layer**

* Core layer contains server Room and server room have firewall, Router, 2 - L3 + switches, patch Rack server inverter, etc

② **Distribution layer**

* Distribution layer contains 4 - switches

* The 4 switches are :-

① Main Build → R7 (Room no 213, 2nd floor)

② Main Build → R1 (Room no - 301, 3rd floor)

③ Main Build → R6 (Room no - 113, Ist floor)

④ Main Build → R2 (Room no - 201, 2nd floor)

This 4 switches contains
① 18 Port Gigabit Easy smart switch with 16 port
TL - SG1218MPE

② SG3428XMP → 24 Port, Gigabit & 4 - Port 10 GE SFP+, L2+ Managed switches with 24 port PoE+

→ Now let's discuss these 2 above components :-

① TP - Link TL - SG1218MPE
* Likely used for : CCTV Camera wifi access point etc
* 18 port Gigabit "Easy smart" Switch
* 16 Port support PoE+

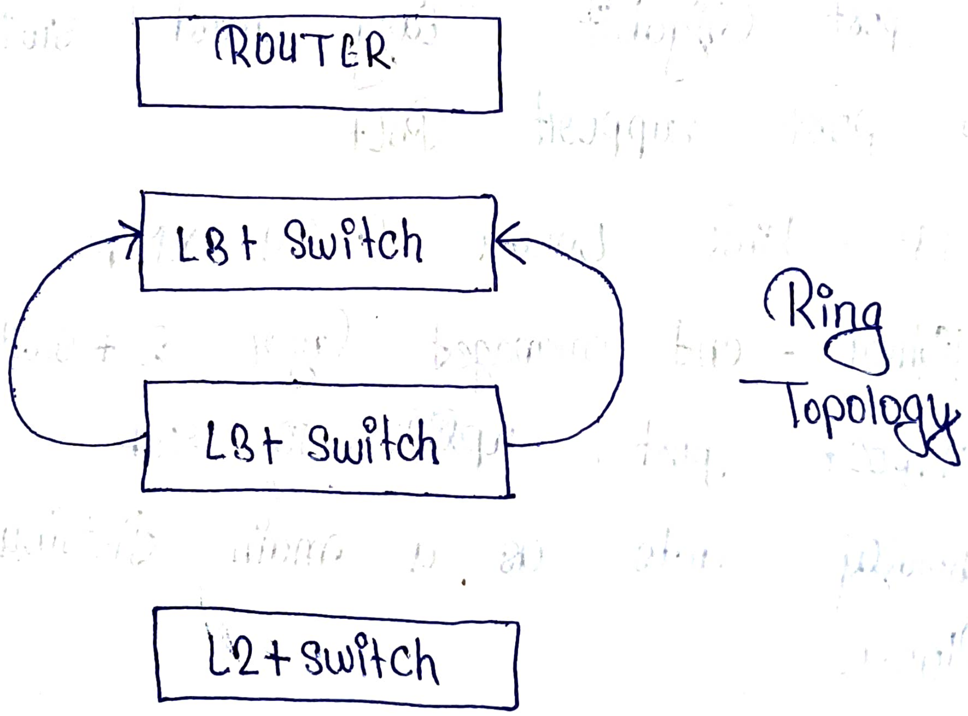
② TP - Link Omada SG3428XMP
* Higher - end managed layer 2 + switches
* 24 PoE+ port, uplink, 10G SFP
* usually acts as a main distribution for each floor.

→ Access layer
* Access layer contain 3 - Switches i.e
1) Main Build R1 - [ROOM no - 301, 3rd Floor]
2) Main Build R4 - [ROOM no - 05, Ground Floor]
3) Main Build R5 - [ROOM no - 18 Ground Floor]

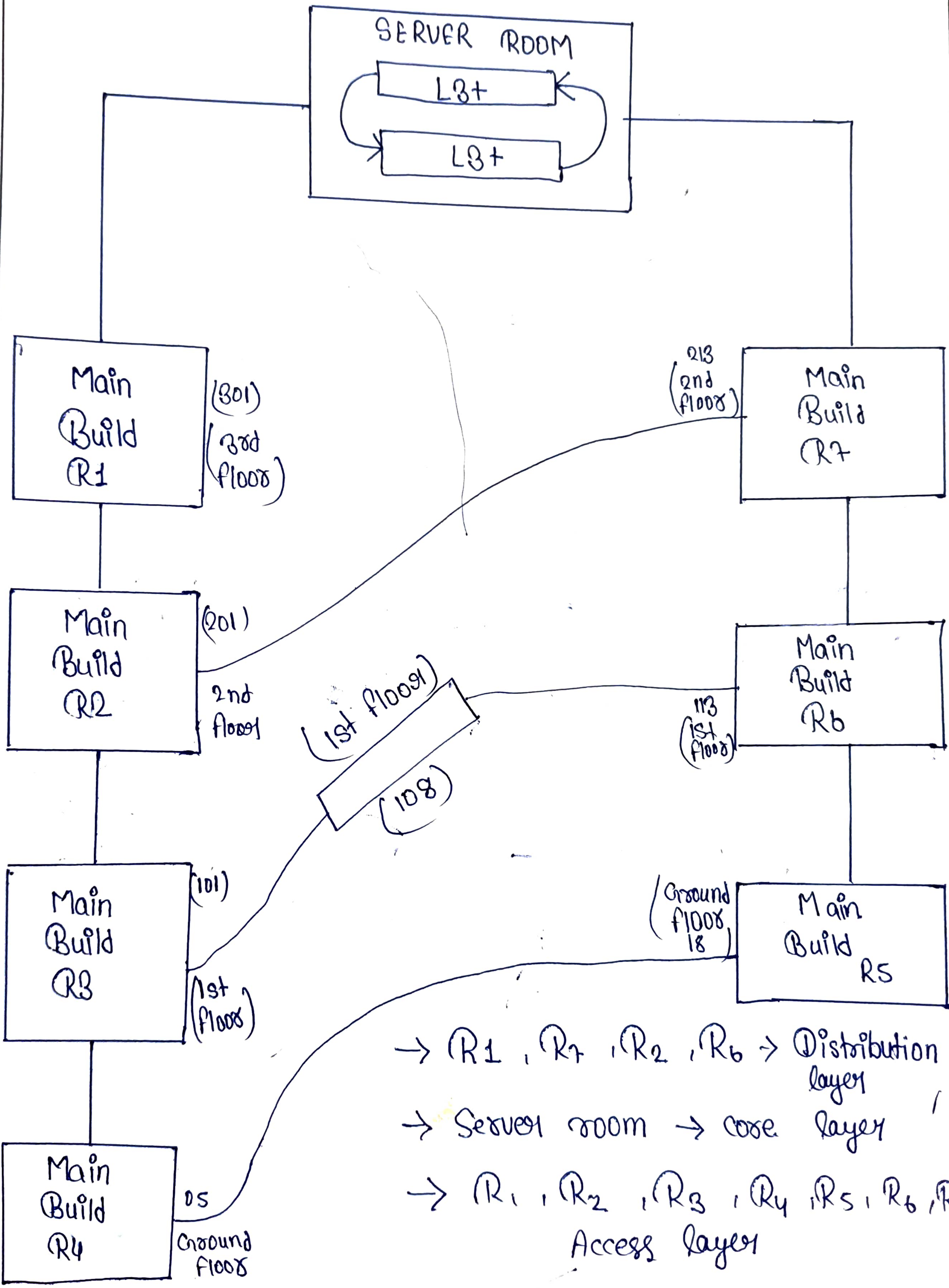
* Every Switches act as access layer which help to provi network connectivity to all classroom Lab faculty Room, etc.

* Access layer directly connect
  1. Ethernet port
  2. Wireless Ap
  3. Lab System
  4. outdoor wireless (AP (EX - 018))

→ Server Room Architecture / topology



Topology



→ R1, R7, R2, R6 → Distribution layer

→ Server room → core layer

→ R1, R2, R3, R4, R5, R6, R7 Access layer

→ Difference b/w wired & wireless speed ?

Wired Network	Wireless Network
* Faster Communication	* Slower communication
* Stable Connection	* Affected by distance /interface
* Low latency	* higer latency
* Dedicated physical medium	* Shared radio medium

→ Ethernet provide direct physical communication using network cables resulting in higher speep & stable performance.

→ Wireless Connection uses radio waves which are affected by interference, distance, obstacles & multiple connected users.

Command Prompt

```
Windows IP Configuration

Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

thernet adapter Ethernet 2:

  Connection-specific DNS Suffix  . :
  Link-local IPv6 Address . . . . . : fe80::992c:6cd2:c438:a76c%8
  IPv4 Address. . . . . . . . . . . : 192.168.56.1
  Subnet Mask . . . . . . . . . . . : 255.255.255.0
  Default Gateway . . . . . . . . . :

reless LAN adapter Local Area Connection* 9:

 Media State . . . . . . . . . . . : Media disconnected
 Connection-specific DNS Suffix  . :

reless LAN adapter Local Area Connection* 10:

Media State . . . . . . . . . . . : Media disconnected
Connection-specific DNS Suffix  . :

eless LAN adapter Wi-Fi:

Connection-specific DNS Suffix  . :
Link-local IPv6 Address . . . . . : fe80::3e82:c0b0:3123:d815%5
IPv4 Address. . . . . . . . . . . : 10.118.32.243
Subnet Mask . . . . . . . . . . . : 255.255.255.0
```

WIRELESS CONNECTION

↳ using "ipconfig"

Command Prompt

```
   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::992c:6cd2:c438:a76c%8
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Local Area Connection* 9:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 10:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::3e82:c0b0:3123:d815%5
   IPv4 Address. . . . . . . . . . . : 10.118.32.243
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 10.118.32.2

C:\Users\ADMIN>
```

WIRED CONNECTION

→ using IPCONFIG

Q) Role of subnet mask in network ?

→ A subnet mask identifiers the network portion & host portion of an IP address. It helps device determines whether another device belongs to the same network or different network.

* It is important for :-

1. Network segment
2. Efficient routing
3. Communication b/w devices

→ OBSERVATION

During the study of the college network infrastructure.

1. A centralized server room was used as network backbone.

2. Two L3 switches were connected in redundant cyclic structure.

3. Managed TP - Link PoE switches were used in network cabinets

→ Advantage of Existing Network ?

* Scalable architecture
* Centralized management.
* Support PoE+ device.
* Easy Explansion of floor & classroom.
* Support both wireless & wired
* Redundant link for failover

CONCLUSION

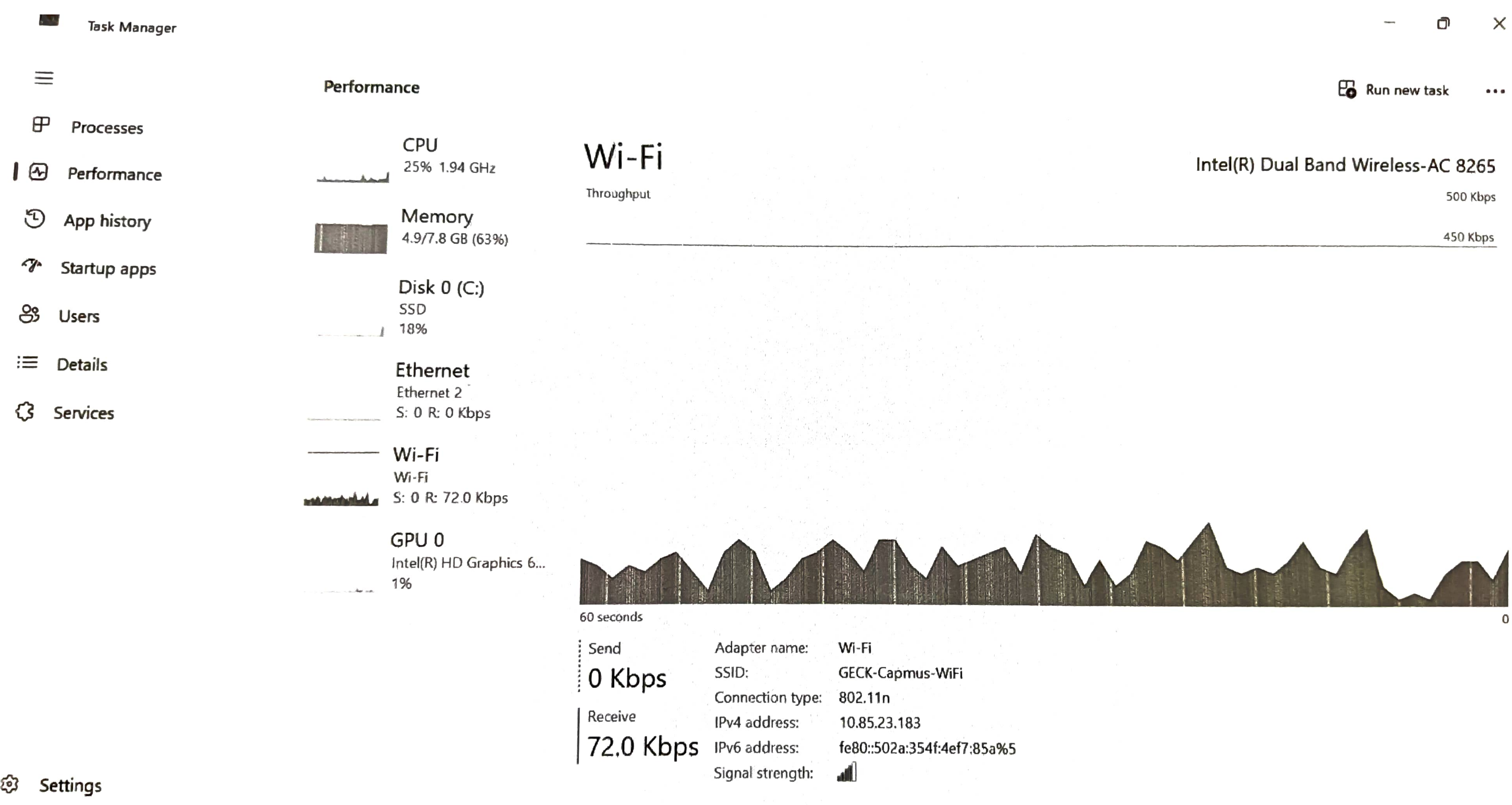
The obsenced campus LAN follow a Hybrid hierarchical topology consist of tree, Extended Star, Ring. The network uses managed PoE+ switch, wireless access Point Ethernet port & centralized serverinfrastructure to provide efficient wired & wireless communication across the campus

Command Prompt

```
Microsoft Windows [Version 10.0.22621.4317]
(c) Microsoft Corporation. All rights reserved.

C:\Users\ADMIN>getmac

Physical Address    Transport Name
=================== ==========================================================
B0-0C-D1-2C-3A-DF   Media disconnected
E6-16-CA-83-CF-DB   \Device\Tcpip_{27635621-3133-4B8B-9255-E7C4BB9E0EA7}
0A-00-27-00-00-08   \Device\Tcpip_{3F28BB86-23C7-4F82-8497-6D44C1776D07}

C:\Users\ADMIN>
```

→ MAC ADDRES

→ using "getmac"


Task Manager
Processes
Performance
App history
Startup apps
Users
Details
Services
Settings
Performance
Run new task
CPU
25% 1.94 GHz
Memory
4.9/7.8 GB (63%)
Disk 0 (C:)
SSD
18%
Ethernet
Ethernet 2
S: 0 R: 0 Kbps
Wi-Fi
Wi-Fi
S: 0 R: 72.0 Kbps
GPU 0
Intel(R) HD Graphics 6...
1%
Wi-Fi
Intel(R) Dual Band Wireless-AC 8265
Throughput
500 Kbps
450 Kbps
60 seconds
0
Send
0 Kbps
Receive
72.0 Kbps
Adapter name: Wi-Fi
SSID: GECK-Capmus-WiFi
Connection type: 802.11n
IPv4 address: 10.85.23.183
IPv6 address: fe80::502a:354f:4ef7:85a%5
Signal strength: